

Priyanka Anil Lakariya

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EDUCATION

Northeastern University | *Master of Science in Robotics* | **GPA: 3.86/4** Sep 2025 - May 2027
Coursework: Robotics Sensing & Navigation, Mobile Robotics, Robot Mechanics & Control, Computer Vision, Reinforcement learning, Advanced Field Robotics

Savitribai Phule Pune University | *Bachelor of Engineering in Mechanical* | **GPA: 3.5/4** Sep 2019 - Jul 2023
Coursework: Basic Electronics Engineering, Basic Electrical Engineering, Electrical & Electronics Engineering, Mechatronics, Measurement Lab, Fluid Power & Control Lab

SKILLS

Programming : Python, C++, Embedded C, SQL, MATLAB, ROS/ROS2
Perception : OpenCV, PCL, YOLO, AprilTag, DBSCAN, PnP, Shape & Pose Estimation, RGB-D Camera, LiDAR
SLAM : RTAB-Map, LIO-SAM, ORB-SLAM3, VSLAM, KF, EKF, Sensor Fusion
Navigation : Nav2, AMCL, DWA, A*, Flood Fill, PID, Model Predictive Control (MPC), Pure Pursuit Control
Learning : PyTorch, TensorFlow, scikit-learn, GAN, LSTM
Simulation : Gazebo, MuJoCo, RViz, Simulink, RoboAnalyzer, SolidWorks
Hardware : IMU/Gyroscope, Motor Control, Sensor Calibration, I2C, Modbus, TCP/IP
Tools : Linux, Git, NumPy, SciPy, Pandas, Matplotlib
Certification : OpenCV Bootcamp, Autonomous Robots (NVIDIA), Industrial Automation

EXPERIENCE

Northeastern University + MIT Sea Grant | *Graduate Research Assistant* Aug 2026 - Present

- Training a YOLOv8 instance-segmentation model on COCO polygon-annotated microscopy data
- Generating synthetic composites to counter severe class imbalance in the training set, improving detection

Innomotics India Pvt. Ltd. (formerly Siemens Large Drive) | *Manufacturing Professional* Jul 2023 - Jul 2025

- Conducted FAI & CAPA; resolved quality issues with suppliers and internal departments with FPY 95-100%
- Coordinated supplier timelines, lead times, and production schedules for **3.3-11KV** medium-voltage motors, scaling monthly output from **10** to **25** units
- Classified **2** years of **1000+** order records into A/B/C material demand using **Advanced Excel**, implemented **Kanban/JIT** to stabilize material supply, and automated **MIS reporting** to cut report cycle time by **40%**
- Tested and validated **3.3-11KV** VFDs against **hardware configurations**, ensuring reliability across assembly

Wissenquelle - OneStop AI | *Robotics Project Intern* Feb 2022 - Apr 2022

- Derived **DH parameters** and computed **forward kinematics** for a 6-DOF PUMA 560, 3-DOF planar manipulator, and 4-DOF SCARA in **RoboAnalyzer**, resolving end-effector poses across joint configurations
- Validated kinematic models against simulation, confirming joint-space to task-space consistency

PROJECTS

Barrett-Style Dexterous Hand | *MuJoCo, ROS2 Jazzy, MoveIt2, URDF/MJCF, SolidWorks* [Link](#)

- Redesigning a rigid 3-finger hand into a Barrett-style dexterous hand, adding a gear-driven finger-spread DOF ($\varnothing 40\text{mm}$ spur gears) and 2-link underactuated fingers, re-actuated on Dynamixel XL330-M288-T
- Mounting the hand on **AgileX Piper** 6-DOF arm and driving it end-to-end from one MoveIt2 configuring SRDF and controllers to execute collision-aware **reach-spread-grasp** sequence

GPS-Denied Drone SLAM for Underground Mine Mapping | *ROS2, LIO-SAM, RTAB-Map, PCL, Gazebo, Nav2* [GitHub](#)

- Integrated LIO-SAM + IMU pipeline to achieve globally consistent 3D reconstruction across cave meshes
- Constructed RTAB loop closure across **10,578 nodes**, generating 3D point cloud of tunnel at **0.01m** voxel resolution
- Deployed frontier-based exploration via Nav2 + OctoMap, achieving 100% waypoint coverage across cave topology

Optical Flow Estimation via Global Matching | *PyTorch, Transformers, Self/Cross-Attention, MPI Sintel, CUDA*

- Built a **GMFlow-style** network - CNN encoder, 2D positional encoding, and **2** shared-weight transformer blocks (self- then cross-attention) feeding a **global softmax matching** layer - regressing dense flow on MPI Sintel at **64×64**
- Reached **0.334 EPE** (best epoch **0.311**) over 50 epochs, down from a **1.17** baseline, supervising final + matching-only flow with a combined EPE loss

Homography Estimation & Panorama Mosaicing | *Python, OpenCV, DLT, SVD, NumPy*

- Implemented homography (**DLT**, SVD solve of $Ah = 0$), in both 4-point and overdetermined least-squares fits
- Warped a source into a target frame by backward-mapped perspective warp to **15.4** Frobenius norm
- Stitched **5** rotation-only photos into a **1016×1223** panorama by chaining per-pair homographies outward from the center frame, compositing overlaps with flat-alpha blending

3D Scene Reconstruction via Structure-from-Motion | *RANSAC, EPnP, Bundle Adjustment, OpenCV*

[GitHub](#)

- Developed a SfM pipeline using SIFT with GrabCut, extracting **1199 keypoints per image** across a 62-image dataset
- Implemented **Lowe's ratio test** and RANSAC estimation to reject outliers, achieving $\geq 70\%$ inlier ratios
- Reconstructed camera trajectory via Essential Matrix decomposition, triangulating a **12,968-point** sparse point cloud
- Achieved **0.83 px mean reprojection error** by sparse bundle adjustment with COO Jacobian, producing an **8,672-point** point cloud in Meshlab

Real-Time Pedestrian Trajectory Prediction | *RTMPose, SMPL, Social-LSTM, CVAE, PyTorch*

[GitHub](#)

- Engineered dual-stream architecture combining **Social-LSTM** with Transformer pose encoder, achieving **35-46%** error reduction on JTA dataset and **0.67m minADE** on ETH/UCY benchmark
- Deployed **RTMPose + SMPL** at **30 FPS**, extracting 17 keypoints to predict pedestrian intent **0.8-1.0s earlier**
- Developed **CVAE decoder** generating K=20 multimodal trajectories over a 4.8s horizon

Autonomous Visual Navigation in Dynamic Environments | *MuJoCo 3.5, Nav2, OpenCV, Python, AMCL, DWA*

[GitHub](#)

- Built a **differential-drive robot** (camera, Lidar) with a **ROS2 Humble** sensor bridge and **Nav2** stack for goal-reaching in a **10×10m arena** with **3 dynamic obstacles**
- Implemented HSV color thresholding for goal detection, stable across **5 Gaussian lighting-noise levels** ($\sigma = 0.0-0.8$)

Kinematic Modeling and Trajectory Tracking for UR5 | *Python, Newton-Raphson IK, Jacobian*

[GitHub](#)

- Derived **forward kinematics** for UR5 from DH parameters, mapping joint angles to end-effector pose in **SE(3)**
- Solved **inverse kinematics** with a **damped least-squares** solver (29-iteration cold-start convergence), tracking circular and figure-eight paths across **157 waypoints** at fixed end-effector orientation

Design and Analysis of Stewart Platform | *SolidWorks, Workspace Analysis, Motion Study*

[Link](#)

- Designed **500mm diameter platforms** and **200mm actuator stroke** (200–400mm range); conducted motion study identifying **50% usable height range**, defining safe zone at 250–350mm height & $\pm 12^\circ$ tilt
- Analyzed workspace geometry achieving **200mm vertical travel**, **± 150 mm lateral displacement**, and **$\pm 15-18^\circ$ tilt capability**; characterized singular boundaries at fully extended and fully retracted positions

Multi-Robot Warehouse Automation - Flipkart GRiD 2023 | *ESP32, Image Processing (OCR), Multi-Robot Coordination*

- Built a multi-robot warehouse system with an **ESP32 master** for pickup, sort, & delivery, for parcel classification
- Built dynamic task allocation, collision avoidance, and floor routing across the fleet

RoboCon 2021 | *Team Anonymous - All-India Rank 16*

- Built a pneumatic actuated roller arrow-gripper mechanism to pick up, throw, and reposition arrows for task
- Engineered a **mecanum/omni-directional drive** under I2C microcontroller control, for swift throw and defence

Low-Cost Stereolithography-Based Resin 3D Printer | *Arduino, I2C, VAT Photopolymerization*

- Built a low-cost **vat-photopolymerization** printer with accuracy at **50-micron** layers, **60%** below SLA cost
- Developed the **I2C** control stack (LCD, TMC2209, UV) and tuned exposure timing, validated on functional prints

RESEARCH

Pose-Guided Trajectory Prediction (PGTP-Net) | *Ongoing*

[Draft](#)

- Extending a dual-stream **Social-LSTM + transformer pose** architecture toward full **3D pose-conditioned** pedestrian forecasting, integrating **SMPL**, **HMR2.0**, and **WHAM** for monocular 3D pose recovery

IMPACT, AWARDS & RECOGNITION

Robotics

- **All-India Rank 16** (Finalist), RoboCon 2021 - Team Anonymous
- Finalist, **Flipkart GRiD** Robotics Challenge 2023
- Quarter-Finalist, **Swadeshi Microprocessor Challenge**, MeitY (Govt. of India), 2021
- Chairman, **MESWCOE Roboclub**